



PROJECT MANAGEMENT CASE STUDY

Rapid Infrastructure for the NHS COVID-19 Response

Standing up an emergency environment at speed, under lockdown restrictions and rising patient demand

Computacenter

Role: Project Manager

Overview

When the COVID-19 pandemic reached the UK, the NHS needed new IT environments and infrastructure stood up in days and weeks rather than months. At Computacenter, I was Project Manager for the rapid design, build, and deployment of this emergency infrastructure. I led a team of engineers across networking, messaging, and Wintel, alongside third-party contractors, to deliver services that clinical and support teams could rely on from day one, while patient numbers rose and national lockdown restrictions changed how we could work.

Scope

- **Core server estate:** Windows servers built from gold images, plus Exchange, SQL and other databases, and WSUS for patching
- **Identity and directory:** Domain controllers and domain services, Active Directory, Group Policy, LDAP
- **Network and security:** Cisco switching, fibre connectivity, WAN and VPN connectivity, firewall servers and policies, SSL, DNS
- **Remote and application access:** RDP, App-V, and secure remote access for staff working away from site
- **Communications:** VoIP telephony and unified communications
- **Hardware:** Rapid procurement, staging, and delivery of servers, network equipment, and end-user devices

The Challenge

- **Extreme urgency.** This was an emergency response. Timelines were measured in days, and services had to be right first time because clinical teams depended on them.
- **Rising patient demand.** Capacity needs grew quickly and unpredictably, so the environment had to scale without redesign.
- **Lockdown restrictions.** Site access was limited and social distancing shaped how engineers could work. Much of the build, configuration, and support had to be done remotely.
- **Supply chain pressure.** Demand for hardware surged worldwide, so sourcing, lead times, and delivery of equipment needed close management.
- **Security and data sensitivity.** Health services handle sensitive data, and speed could not come at the cost of security, particularly with more remote access.
- **Coordination across teams.** Networking, messaging, Wintel, contractors, and NHS stakeholders had to operate as one team, largely without being in the same room.

Approach

Rapid mobilisation and clear priorities. I formed the delivery team quickly, agreed a short list of critical services with NHS stakeholders, and sequenced work by clinical impact. Scope was kept tight, with anything non-essential deferred so the team could focus on what mattered most.

Standardisation for speed. Server gold builds and consistent Group Policy allowed infrastructure to be deployed quickly and repeatably, with far fewer configuration errors. Standard designs also made it easy to

replicate the environment as demand grew.

Building the foundations first. Network, AD, and DNS were prioritised because everything else depends on them. Getting identity and connectivity right early allowed messaging, databases, telephony, and application services to be built in parallel.

Remote-first delivery. Under lockdown, engineers configured and tested services remotely wherever possible, and on-site visits were limited to essential hardware installation with safe working practices. Daily stand-ups and shared trackers kept a distributed team aligned.

Managing the supply chain. I worked closely with suppliers on hardware availability and delivery, tracked assets from the moment they were ordered, and held spare gold stock where possible to allow quick swap-outs if devices arrived faulty or demand jumped.

Security built in. Firewall policies, VPN access, and SSL were reviewed and tested before services went live, and access was controlled through Active Directory and Group Policy so remote working did not weaken the security position.

Testing and handover under pressure. Compressed but disciplined end-to-end testing validated that identity, network, messaging, and voice worked together before services were handed to operational teams, with clear escalation paths for early-life support.

Fast, honest communication. Short daily updates to stakeholders covered progress, risks, and decisions needed. Escalating issues early kept decisions fast and stopped small problems becoming delays.

Results

- Emergency infrastructure delivered at pace to meet the urgent needs of the NHS response
- Services built on a standardised, repeatable design that could scale as demand increased
- Delivery maintained under lockdown restrictions through remote-first working and safe on-site practices
- Hardware supply risks contained through early ordering, asset tracking, and spare stock
- Security standards maintained despite the speed of delivery and increase in remote access

Key Lessons

- 1 Prioritise ruthlessly.** In an emergency, a short list of critical services delivered well beats a long list delivered late.
- 2 Standardise to go faster.** Gold builds and consistent policies made deployment repeatable and scaling straightforward.
- 3 Foundations decide the schedule.** Network, directory, and DNS had to be right before anything built on them.
- 4 Design for remote delivery.** Working remotely kept the project moving when access was restricted, and it is now a proven way of delivering.
- 5 Plan for hardware failure.** Spare stock and asset tracking protected the schedule from supply chain disruption.
- 6 Communicate early and plainly.** Short, frequent updates kept a distributed team and its stakeholders aligned.

Reflection

This project showed me that delivery under crisis conditions depends on clear priorities, strong standards, and calm, honest communication. Knowing the work supported frontline NHS teams gave the whole team a shared sense of purpose.